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INSID&S Lab
*Interaction and Semantics for
Innovation with Data & Services*

2.1- Part I

Knowledge Graphs & RDF

Knowledge Graphs & RDF: Outline

■ Knowledge Graphs

- What they are, intuitively
- Why they are useful
- What they are, precisely

Part I – This presentation

■ The RDF data model

- RDF

Part II

Knowledge Graphs

What they are, intuitively

Why they are useful

What they are, precisely

Representing Knowledge with *Knowledge Graphs*

Why (knowledge) graphs?

Slide by
A. Bordes (Facebook)
E. Gabrilovich (Google)

- We're surrounded by **entities**, which are connected by **relations**
- We need to store them somehow, e.g., using a **DB** or a **graph**
- **Graphs** can be processed **efficiently** and offer a convenient **abstraction**

At the Convergence of More Disciplines

- Data Management

- ☐ Databases and especially graph databases
- ☐ I assume you know it

- Artificial Intelligence

- ☐ Knowledge bases

- Web technologies

- ☐ Making data accessible on the web

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Reasoning and Inference

rea·son·ing | \ 'rēz-niŋ , 'rē-zən-iŋ \

Definition of *reasoning*

1 : the use of reason

especially : the drawing of inferences or conclusions through the use of reason

2 : an instance of the use of reason : ARGUMENT

■ Inferences / conclusions

- An inference establishes a relation

- Premises → Conclusion

- Computational models of inference

- Which conclusions can we draw from a set of premises?

- Why is the conclusion correct?

Reasoning and Inferences

1. Madonna is a pianist or a singer.
If Madonna is a pianist, then she is a musician.
Madonna is not a musician.
Therefore: Madonna is a singer.
2. Obama is a gardener.
All gardeners are Sicilians.
Therefore: Obama is Sicilian.

Correct?
Y | N



Obama gardening



Sicilian Gardens

Reasoning and Inferences

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Why correct?



Obama gardening



Sicilian Gardens

Reasoning and Inferences

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- Obama **is** a gardener.
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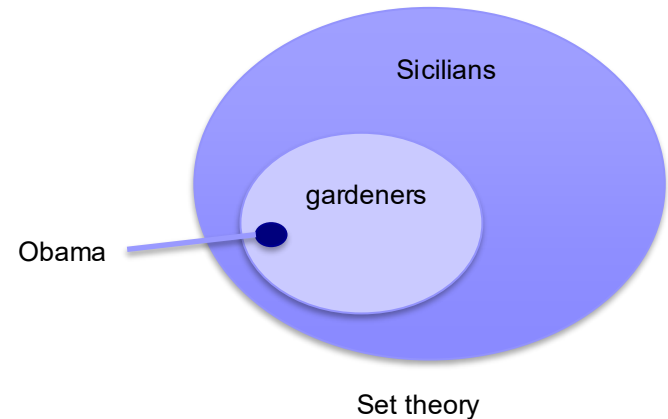
#1.a) Agreement on the meaning of
connectives between prepositions
 (e.g., **AND**, **OR**, **NOT**, **IF/THEN**)

α	β	α OR β
0	0	0
0	1	1
1	0	1
1	1	1

α = Madonna is a pianist
 β = Madonna is a singer

#1.b) If the **premises** are true,
 the **consequences** **must** be true as well

Deduction



Knowledge Bases (KBs) #1

- Goal: model agents that can manipulate knowledge and execute knowledge-intensive tasks
 - Facts + general domain knowledge
 - {Paris is located in France, France is located in Europe} + {is located in is a transitive relation} $\subseteq$ KB
 - Reasoning: inferences and inference-aware queries
 - $\text{KB} \vdash \{\text{Paris is located in Europe}\}$
 - Where is Paris located? {France, Europe}
 - ... more on this when we introduce ontologies ☺

Knowledge Bases (KBs) #2

- Goal: model agents that can manipulate knowledge and execute knowledge-intensive tasks
 - Difference with databases
 - In KBs, knowledge = persisted knowledge + inferences
 - In KBs, data organization inspired by human conceptualizations; inferences inspired by “human” reasoning
 - Similarity to databases
 - CRUD operations: create, read, update, e delete
 - Persistence and localization
 - Reliability of query processing (+ reasoning in KBs)
 - based on well-understood semantics, explainable and traceable (step-by-step)

Knowledge Graphs: KBs with Specific Features

■ Specific features

- ☐ Binary relations
- ☐ Data management for AI
- ☐ Mix well with LLMs and generative AI
- ☐ Mix well with other approaches based on NNs

Knowledge Graphs

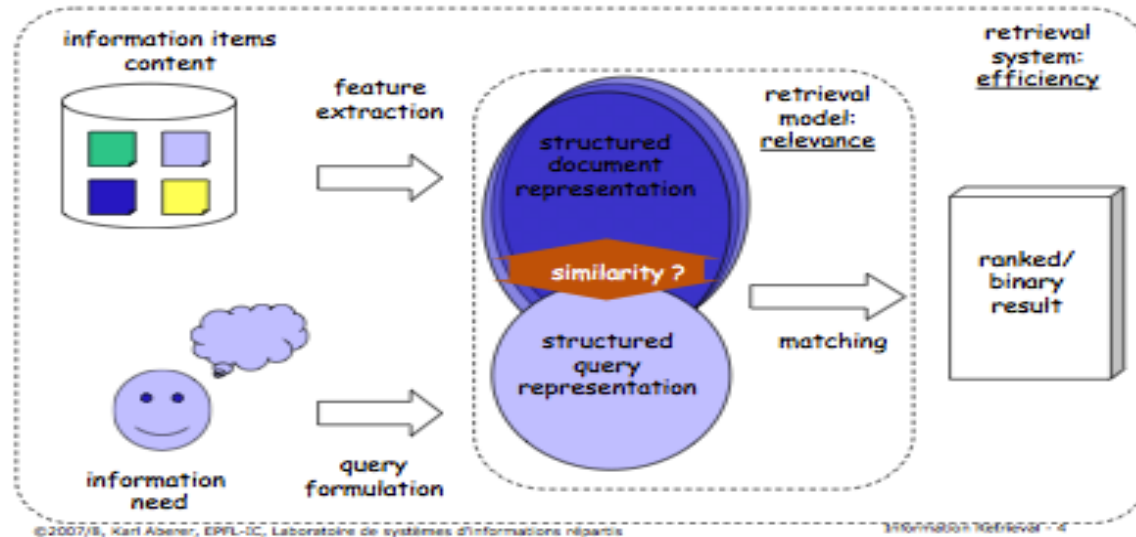
What they are, intuitively

Why they are useful

What they are, precisely

Data on the Web Documents vs. Factual Information

Traditional Keyword-based Search



- Model to evaluate the similarity between a query (set of keywords) and a document (set of words)
 - Vector-Space Model
 - Query: a vector of words / Document: a vector of words

Document Representation in a Traditional IR System

■ Representation of documents:

□ Term-document Matrix

- $\langle i, j \rangle = 1$ if the i -th term occurs in the j -th document, 0 otherwise
- OR $\langle i, j \rangle =$ a weight = how relevant the i -th term is for the j -th document
 - E.g., TF-IDF (Term Frequency – Inverse Document Frequency), as in the image

Index →	0	1	2	3	4	5	6	7	8	
Documents ↓	and	document	first	is	one	second	the	third	this	← terms
"This is the first document."	0	0.46979139	0.58028582	0.38408524	0	0	0.38408524	0	0.38408524	
"This document is the second document."	0	0.6876236	0	0.28108867	0	0.53864762	0.28108867	0	0.28108867	
"And this is the third one."	0.51184851	0	0	0.26710379	0.51184851	0	0.26710379	0.51184851	0.26710379	
"Is this the first document?"	0	0.46979139	0.58028582	0.38408524	0	0	0.38408524	0	0.38408524	

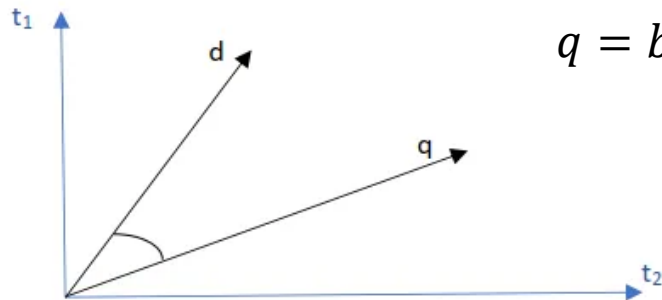
Source: <https://zilliz.com/learn/tf-idf-understanding-term-frequency-inverse-document-frequency-in-nlp>

The Vector Space Model

$$d = a_1, \dots, a_n$$

$$q = b_1, \dots, b_n$$

$$\text{Cosine: } \text{Sim}(d, q) = \frac{d \cdot q}{|d| \cdot |q|} = \frac{\sum_i (a_i \times b_i)}{\sqrt{\sum_i a_i^2 \times \sum_i b_i^2}}$$



a_i and b_i

- correspond to the weight associated with the i -th term t_i (respectively in the document and the query)
- are included in $(0,1)$

Vector-space Retrieval Example

■ Example of Cosine Similarity for two input queries

Query	"This is the first document."	"And this is the third one."
Q1: "first document"	0.7466	0.0000
Q2: "third document"	0.2528	0.4315

Index →	0	1	2	3	4	5	6	7	8	
Documents ↓	and	document	first	is	one	second	the	third	this	← terms
"This is the first document."	0	0.46979139	0.58028582	0.38408524	0	0	0.38408524	0	0.38408524	
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Documents vs Facts

- Documents in IR
- Facts in KB

What's the key difference in their semantics?
What can you do based on the different semantics?

Documents vs Facts

■ Documents in IR

- ☐ Can be relevant or not (for a specific information need)
- ☐ Searches, clustering, etc.

■ Facts in KB

- ☐ Can be true or false (or partially... etc.)
- ☐ Queries and inferences

At the Convergence of More Disciplines

■ Data Management

- Databases and especially graph databases
- I assume you know it

■ Artificial Intelligence

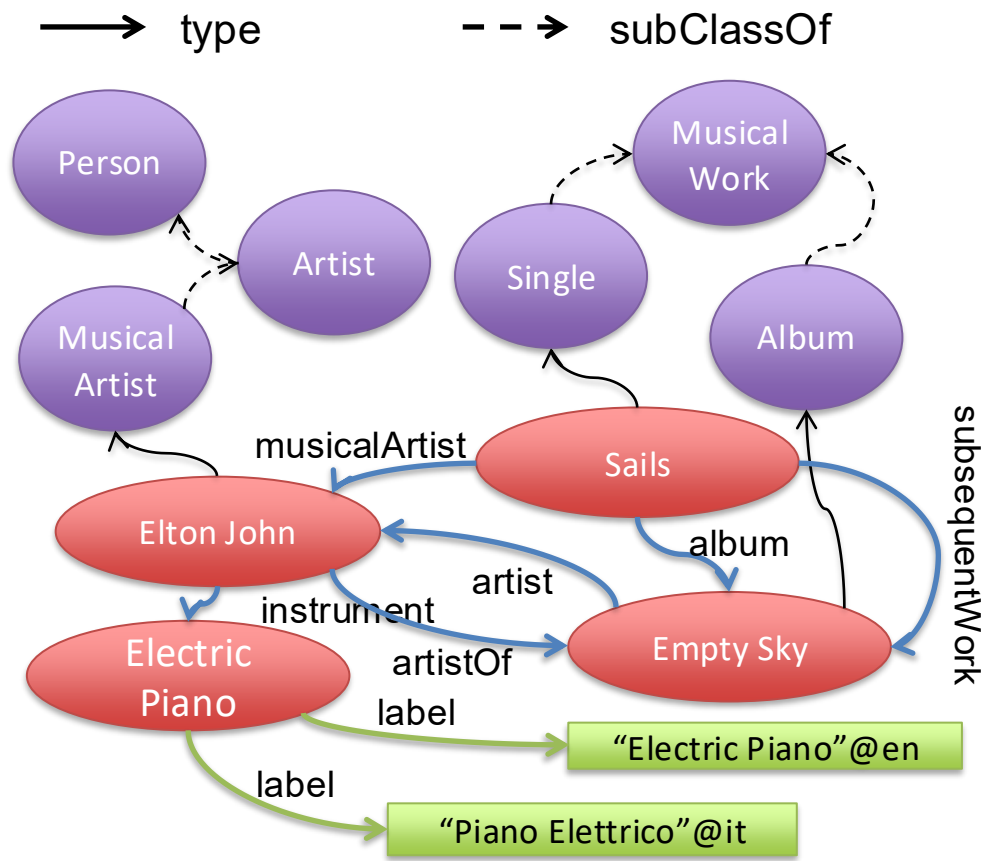
- Knowledge bases

■ Web technologies

- Making data accessible on the web

Knowledge Graphs (Facts vs. Graph View)

<Electric Piano, label, "Electric Piano"@en>
<Electric Piano, label, "Piano Elettrico"@it>
<Elton John, instrument, Electric Piano>
<Empty Sky, artist, Elton John>
<Sails, musicalArtist, Elton John>
<Sails, album, Empty Sky>
<Elton John, instrument, Electric Piano>
<Elton John, type, Musical Artist>
<Sails, type, Single>
<Empty Sky, type, Album>
<Musical Artist, subClassOf, Artist>
<Artist, subClassOf, Person>
<Album, subClassOf, Musical Work>
<Single, subClassOf, Musical Work>
<Elton John, artistOf, Empty Sky>
<Sails, subsequentWork, Empty Sky>



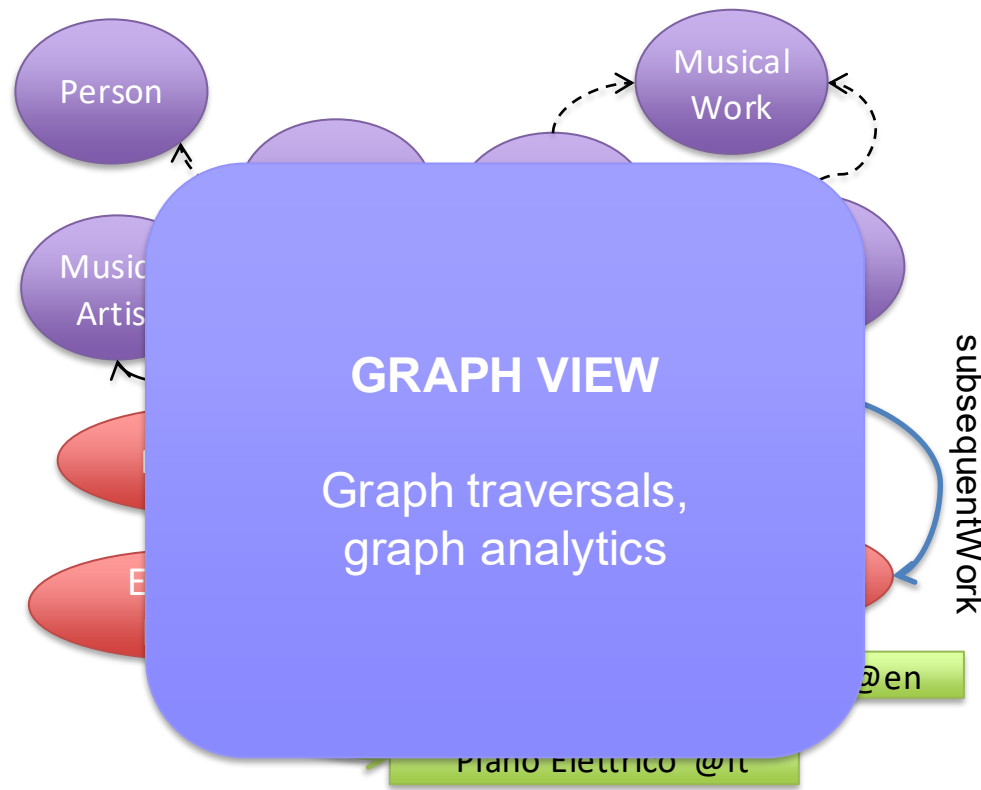
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 <Elton John, instrument, Electric Piano>
 <Empty Sky, artist, Elton John>
 <Sails, musicalArtist, Elton John>

FACT VIEW

Expressive queries,
Data transformations, logic-
based inferences

→ type - - → subClassOf



Two Open Knowledge Graphs

Automatically
extracted from
Wikipedia



www.dbpedia.org
live.dbpedia.org

Are you looking for the New DBpedia Page? Please browse this way --> www.dbpedia.org

Data Download

[Browse the DBpedia Datasets](#)

[Go to Latest Release](#)

www.wikidata.org

Welcome to Wikidata

the **free** knowledge base with **92,845,639** data items that **anyone can edit**.

[Introduction](#) • [Project Chat](#) • [Community Portal](#) • [Help](#)

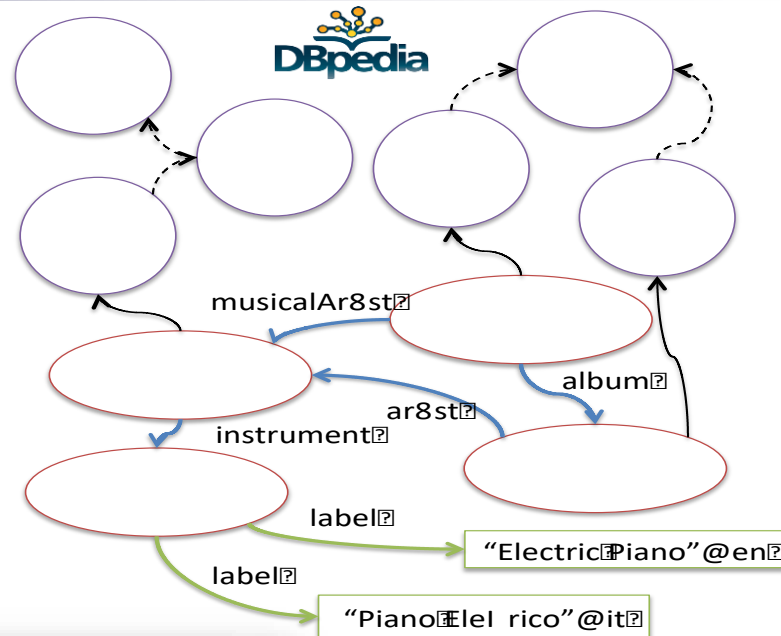
Want to help translate? [Translate the missing messages.](#)

Collaboratively built
like Wikipedia

“Big” Knowledge Graphs

- DBpedia (2023)
 - 4.8M **entities**
- Ontology (schema)
 - 768 **classes**
 - ~3000 **properties**

Class	Instances
Resource (overall)	4,828,418
Place	967,491
Person	1,592,912
Work	552,115
Species	190,369
Organisation	317,867
Other	1,207,664



```

<ElectricPiano, label, ?ElectricPiano"@en>
<ElectricPiano, label, ?PianoElel rico"@it>
<EltonJohn, instrument, ElectricPiano>
<EmptySky, ar8st, EltonJohn>
<Sails, musicalAr8st, EltonJohn>
<Sails, album, EmptySky>
<EltonJohn, instrument, ElectricPiano>
<EltonJohn, type, MusicalAr8st>
<Sails, type, Single>
<EmptySky, type, Album>
    
```

```

<MusicalAr8st, subclassOf, Ar8st>
<Ar8st, subclassOf, Person>
<Album, subclassOf, MusicalWork>
<Single, subclassOf, MusicalWork>
    
```

Knowledge Graphs in the Industry

DOI:10.1145/3331166



Article development led by [acmqueue](https://queue.acm.org)
queue.acm.org

**Five diverse technology companies
show how it's done.**

BY NATASHA NOY, YUQING GAO, ANSHU JAIN,
ANANT NARAYANAN, ALAN PATTERSON, AND JAMIE TAYLOR

Industry-Scale Knowledge Graphs: Lessons and Challenges

*Natasha Noy, Yuqing Gao, Anshu Jain, Anant Narayanan,
Alan Patterson, Jamie Taylor. Communications of the
ACM, August 2019, Vol. 62 No. 8, Pages 36-43*

Common characteristics of the knowledge graphs.

	Data model	Size of the graph	Development stage
Microsoft	The types of entities, relations, and attributes in the graph are defined in an ontology.	~2 billion primary entities, ~55 billion facts	Actively used in products
Google	Strongly typed entities, relations with domain and range inference	1 billion entities, 70 billion assertions	Actively used in products
Facebook	All of the attributes and relations are structured and strongly typed, and optionally indexed to enable efficient retrieval, search, and traversal.	~50 million primary entities, ~500 million assertions	Actively used in products
eBay	Entities and relation, well-structured and strongly typed	Expect around 100 million products, >1 billion triples	Early stages of development and deployment
IBM	Entities and relations with evidence information associated with them.	Various sizes. Proven on scales documents >100 million, relationships >5 billion, entities >100 million	Actively used in products and by clients



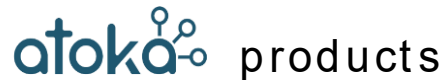
SPAZIO**DATI** WHO WE ARE



BIG DATA COMPANY SPECIALIZED IN BUSINESS KNOWLEDGE

. 2 locations in Italy: Trento, Pisa

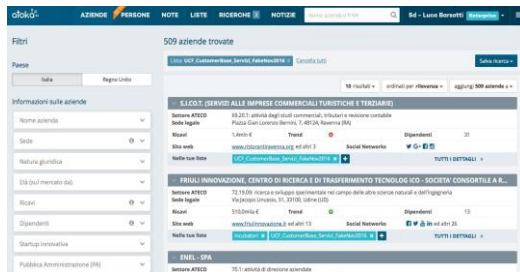
. 25 Sw Eng. & Data Scientists (~30 yrs avg)
. > 100 man-yrs in R&D



A single point of access to company data for lead generation / sales and marketing intelligence.

ATOKA WEB APPLICATION

<https://atoka.io/>



FEATURES:

- ⑩ COMPANY SEARCH & DETAILS
 - JOB ADS
 - GOVERNMENT CONTRACTS
- ⑩ CONTACTS
- ⑩ LISTS
- ⑩ SIMILARITY
- ⑩ NEWS SEARCH
- MONITORING

ATOKA ENTERPRISE EDITION

POSSIBLE CUSTOMIZATIONS:

- BYOD & SOFT INTEGRATION

ATOKA API

(programmatic access to all of the above)

Knowledge Graphs

What they are, intuitively

Why they are useful

What they are, precisely

Features & Impact on Applications

Impact of these features depend on applications and technical implementation choices (e.g., RDF vs. others)

- Factuality and reasoning: question answering
- Natural language friendly representations: human (and NLP) interpretability
- Graph structure: connectivity
- Interoperability: simple triple-based model + standard formats + shared vocabularies
- Data integration: linking and enriching data

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Application 1. Web search

Search results for **jiawei han**

About 836,000 results (0.58 seconds)

hanj.cs.illinois.edu ▾

Jiawei Han

Xiang Ren and **Jiawei Han**, Mining Structures of Factual Knowledge from Text: An Effort-Light Approach, Morgan & Claypool Publishers, 2018 (Series: Synthesis ...)

You've visited this page many times. Last visit: 12/13/19

scholar.google.com ▸ citations ▾

Jiawei Han - Google Scholar

J Han, J Pei, B Mortazavi-Asl, H Pinto, Q Chen, U Dayal, M Hsu. proceedings of the 17th international conference on data engineering, 215-224, 2001.

en.wikipedia.org ▸ wiki ▸ Jiawei_Han ▾

Jiawei Han - Wikipedia

Jiawei Han (Chinese: 韩家炜; born August 10, 1949) is a Chinese-American computer scientist who currently holds the position of Abel Bliss Professor in the Department of Computer Science at the University of Illinois at Urbana-Champaign.



Jiawei Han

Computer scientist

Jiawei Han is a Chinese-American computer scientist who currently holds the position of Abel Bliss Professor in the Department of Computer Science at the University of Illinois at Urbana-Champaign. [Wikipedia](#)

Born: August 10, 1949 (age 71 years), Shanghai, China

h-index: 177

Doctoral advisor: Larry Travis

Notable student: Osmar R. Zaiane

Education: University of Wisconsin-Madison (1985), University of Science and Technology of China

Books [View 5+ more](#)

- Mining Heterogeneous Information Networks
- Outlier Detection for Temporal Data
- Multidimensional Association Analysis
- Mining Structures of Factual Knowledge from Text

Slide by Luna Dong (Amazon). From: [Lockard, Shiralkar, Dong & Hajishirzi: Multi-modal Information Extraction from Text, Semi-structured, and Tabular Data on the Web, tutorial at KDD2020](#)

Search + Browsing

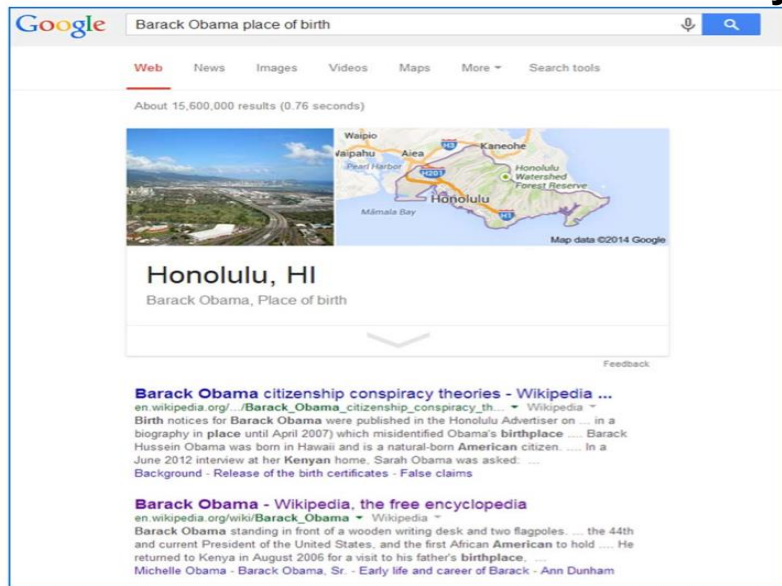
The screenshot shows a Google search for "New York". The search bar at the top contains "New York" and the Google logo. Below the search bar, there are tabs for "Web", "Images", "News", "Maps", "Videos", and "More". The search results are displayed below the tabs. The first result is "New York - Wikipedia, the free encyclopedia" with a link to "en.wikipedia.org/wiki/New_York". The second result is "New York City - Wikipedia, the free encyclopedia" with a link to "en.wikipedia.org/wiki/New_York_City". The third result is "The Official New York City Guide to NYC Attractions, Dining ..." with a link to "www.nycgo.com". The fourth result is "The New York Times - Breaking News, World News ..." with a link to "www.nytimes.com". The fifth result is "New York Magazine -- NYC Guide to Restaurants, Fashion ..." with a link to "nymag.com". The sixth result is "News for new york" with a link to "New York, Responding to Surge of Child Migrants, Forms ...". The seventh result is "NewYork.com - Your Official Site for Travelling To and Living ..." with a link to "www.newyork.com".

Augmenting the presentation
with relevant facts

Slide by
A. Bordes (Facebook)
E. Gabrilovich (Google)

Application 2: Question Answering & Conversational Agents

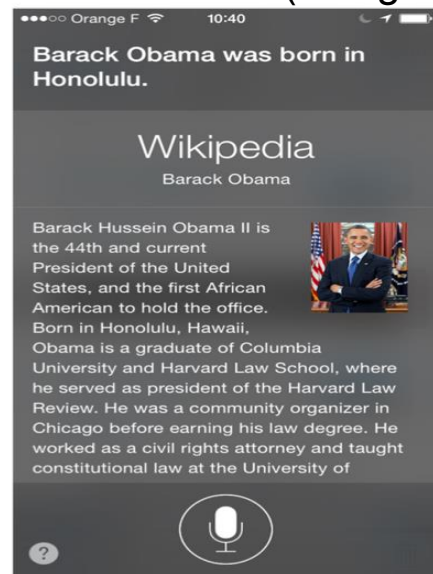
Slide by A. Bordes (Facebook) & E. Gabrilovich (Google)



Google



EVI
(Amazon)



Siri
(Apple)

Talk To Your Graph



Conversational AI for Knowledge Graphs:

Ontology Aware LLM

Natural language queries with SPARQL precision.

From RAG to GraphRAG

Multi-Method Querying:

Combine SPARQL, vector similarity, and full-text search for dynamic data retrieval.

Customizable, Intelligent Agents:

Combine different retrieval methods to deliver conversational insights and improve explainability

The screenshot shows a chat interface with a light blue header and a white body. It contains three question-and-answer pairs. Each question is in a light blue box, and each answer is in a white box with a small icon of a person speaking. The first question is 'What countries are represented in the data?' and the answer is 'The countries represented in the data are Austria, France, Italy, and Norway.' The second question is 'How many buyers from France do you have in the data?' and the answer is 'There are 4,068 buyers from France in the data.' The third question is 'What is the total amount of the tenders from french buyer for the month of march 2025?' and the answer is 'The total amount of the tenders from French buyers for the month of March 2025 is 719,383,824.2.' Below the third answer is a 'Hint' box that says 'Hint: you can also ask the model, for example' followed by a button 'How did you derive this answer?'. At the bottom, there is a box titled 'Called SPARQL: Direct query' containing a SPARQL query:

```
SELECT (SUM(?amount) AS ?totalAmount) WHERE { ?tender a :Tender; :buyer ?buyer; :amount ?amount; :date ?date . ?buyer a :country "France" . FILTER (MONTH(?date) = 3 && YEAR(?date) = 2025) }
```

What countries are represented in the data?

The countries represented in the data are Austria, France, Italy, and Norway.

How many buyers from France do you have in the data?

There are 4,068 buyers from France in the data.

What is the total amount of the tenders from french buyer for the month of march 2025?

The total amount of the tenders from French buyers for the month of March 2025 is 719,383,824.2.

Hint: you can also ask the model, for example [How did you derive this answer?](#)

Called SPARQL: Direct query

```
SELECT (SUM(?amount) AS ?totalAmount) WHERE { ?tender a :Tender; :buyer ?buyer; :amount ?amount; :date ?date . ?buyer a :country "France" . FILTER (MONTH(?date) = 3 && YEAR(?date) = 2025) }
```


Application 3: Recommendation



Slide by Luna Dong (Amazon). From: [Lockard, Shiralkar, Dong & Hajishirzi: Multi-modal Information Extraction from Text, Semi-structured, and Tabular Data on the Web, tutorial at KDD2020](#)

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Rich Snippets in Search Results

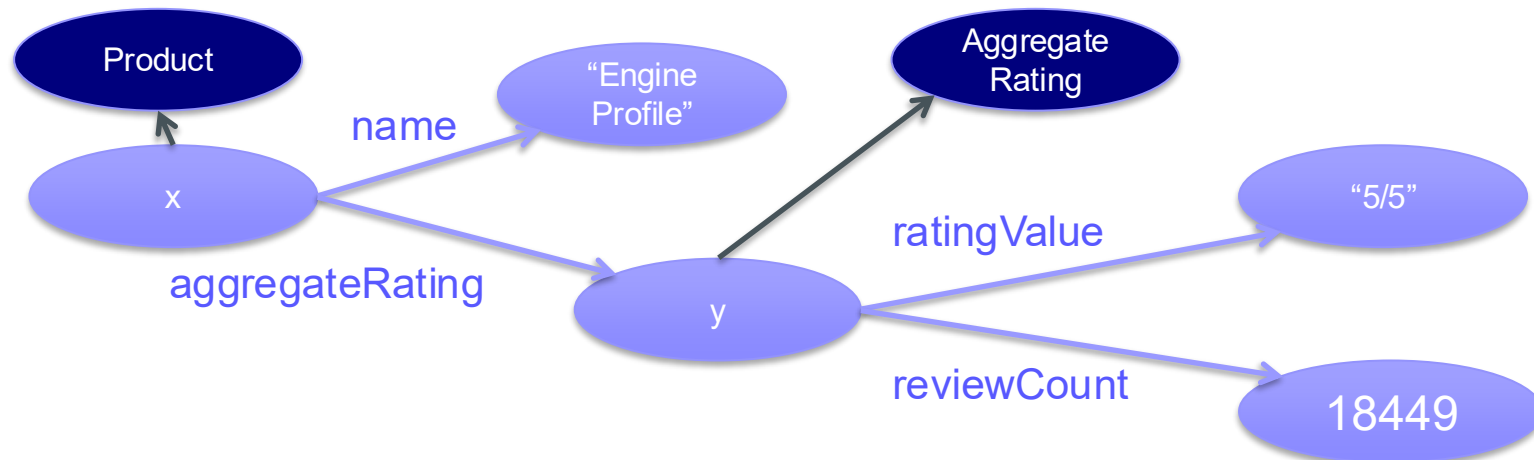


Also useful for
Search Engine Optimization

```
<span itemscope itemtype="http://schema.org/Product">
<span itemprop="name"><strong>Engine Profiles</strong></span>
<span itemprop="aggregateRating" itemscope
itemtype="http://schema.org/AggregateRating">
rated <span itemprop="ratingValue">5</span> / 5<br />
based on <span itemprop="reviewCount">18449</span></span>
```

Engine Profiles
rated **5 / 5**
based on **18449**

Rich Snippets in Search Results



```
<span itemscope itemtype="http://schema.org/Product">
<span itemprop="name"><strong>Engine Profiles</strong></span>
<span itemprop="aggregateRating" itemscope
    itemtype="http://schema.org/AggregateRating">
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based on <span itemprop="reviewCount">18449</span></span>
```

Engine Profiles
rated **5 / 5**
based on **18449**

Semantic Data Integration

Annotations: data linking

BROWSE BY COUNTRY FRANCE 8

ENTITY	JURISDICTION	LINKED TO	DATA FROM
OBOLT GLOBAL LTD	British Virgin Islands	France	Panama Papers
FAIRWINDS SAILING S.A.	British Virgin Islands	France	Panama Papers
SIENNA HOLDINGS INC.	Panama	France	Panama Papers
PROCESOS E INGENIERIA S.A.	Panama	France	Panama Papers
WEBCO INTERNATIONAL S.A.	Bahamas	France	Panama Papers
INTERPETROLEUM LTD.	Bahamas	France	Panama Papers
UNICONSTRUCT INC.	Panama	France	Panama Papers
MADINA AL MINWORA AGENCY CORPORATION	Panama	France	Panama Papers
CARLTON PROPERTIES LIMITED	Bahamas	France	Panama Papers
BELROS HOLDINGS LTD.	Bahamas	France	Panama Papers

FRANCE - PANAMA

France investigating hundreds for tax fraud due to Panama Papers

French magistrates are investigating 26 individuals or institutions suspected of hiding huge sums of money in tax havens and tax authorities are looking into 416 suspected tax dodgers in an inquiry sparked by last year's Panama Papers revelations, according to reports.

Issued on: 05/04/2017 - 15:49

PANAMA PAPERS

« Panama papers » : le business offshore du Crédit agricole et de la BNP

Les « Panama papers » mettent en évidence les pratiques opaques des deux grandes banques françaises dans les paradis fiscaux.

Par Anne Michel, Jérémie Baruch et Maxime Vaudano

EVASION FISCALE

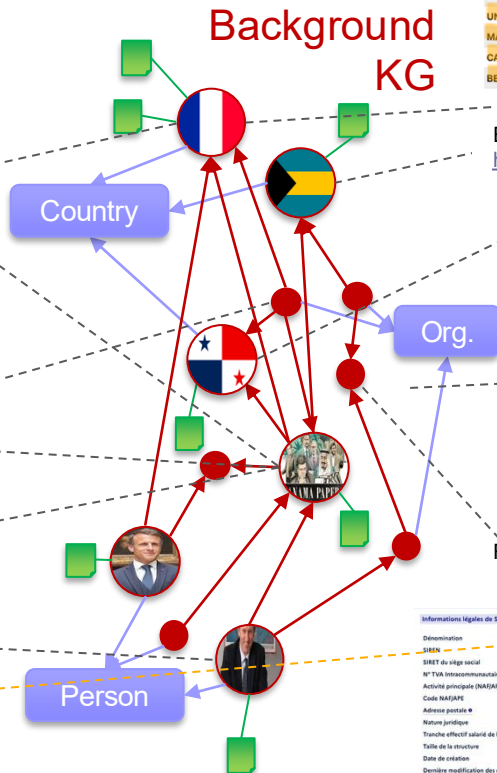
« Panama Papers » : les mauvaises affaires offshore de l'industriel Gérard Autajon

En plus d'un redressement fiscal, d'une amende et d'une peine de prison avec sursis, le PDG a dû faire face à des frais non anticipés sur deux sociétés.

Par Jérémie Baruch

Publié le 21 juin 2018 à 10h54, modifié le 22 juin 2018 à 07h12 - Lecture 3 min.

... Sienna ...



Entities in OffshoreLeaks linked to France

<https://offshoreleaks.icij.org/search?c=FRA&cat=0>

Foundations firms 'offshore' customers through banks

Headquarters	Bank	Number of foundations
Luxembourg	Experta Corporate & Trust Services (100% subsidiary of BGL)	1,659
Luxembourg	Banque J. Safra Sarasin - Luxembourg S.A.	963
United Kingdom	Street Suisse Channel Islands Limited	918
Monaco	HSBC Private Bank (Monaco) S.A.	778
Switzerland	HSBC Private Bank (Switzerland) S.A.	733
Switzerland	UBS AG (subsidiary Rue du Rhône in Geneva)	579
United Kingdom	Credit & Co Trustee (Jersey) Limited	487
Jersey	Bank of Jersey	465
Luxembourg	Société Générale Bank & Trust Luxembourg	404
Luxembourg	Landbank Luxembourg S.A.	378
United Kingdom	Bank of Jersey	378
Spain	Banco Santander	119
Spain	BBVA	19

Foundations firms 'offshore' customers through banks in Wikipedia

Informations légales de SIENNA REAL ESTATE HOLDING FRANCE

Dénomination	492 220 553
SIENNA	492 220 553 00027
SIRET du siège social	N° TVA Intracommunautaire
Activité principale (NAIS/NAPE)	Activité de placement et entités financières similaires
Code NAF/NAPE	64 30Z
Adresse postale	18 RUE DE COURCELLES, 75008 PARIS 8
Nature juridique	SAS, société par actions simplifiée
Taux effectif d'impôt de la structure	Unité non employeur
Taille de la structure	Relevé au Répertoire des Entreprises (RNE), en 2018
Date de création	20/09/2008
Dernière modification des données Insee	06/09/2022

Company data from .data.gouv.fr
<https://annuaire-entreprises.data.gouv.fr/entreprise/sienna-real-estate-holding-france-492220553>

Named Entity Recognition

Named Entity Linking

NIL Prediction

Clustering

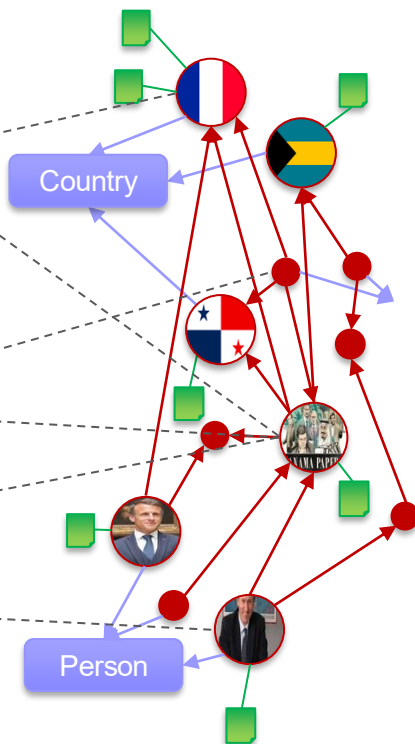
Semantic Annotations vs Data Integration

Annotations: data linking

FRANCE - PANAMA
France investigating hundreds for tax fraud due to Panama Papers
French magistrates are investigating 26 individuals or institutions suspected of hiding huge sums of money in tax havens and tax authorities are looking into 416 suspected tax dodgers in an inquiry sparked by last year's Panama Papers revelations, according to reports.
Issued on: 05/04/2017 - 15:49

PANAMA PAPERS
« **Panama papers** » : le business offshore du **Crédit agricole** et de la **BNP**
Les « Panama papers » mettent en évidence les pratiques opaques des deux grandes banques françaises dans les paradis fiscaux.
Par Anne Michel, Jérémie Baruch et Maxime Vaudano

EVASION FISCALE
« **Panama Papers** » : les mauvaises affaires offshore de l'industriel **Gérard Autajon**
En plus d'un redressement fiscal, d'une amende et d'une peine de prison avec sursis, le PDG a dû faire face à des frais non anticipés sur deux sociétés.
Par Jérémie Baruch
... Sienna ...
Publié le 21 juin 2018 à 10h54, modifié le 22 juin 2018 à 07h12 - Lecture 3 min.



Information extraction and annotation

■ “extraction”

- Sets of entities, relations, etc.

■ “annotations”

- Links between specific sequences of characters in the text and the extracted content
 - E.g., JSON representing the links

Semantic Annotations vs Data Integration

Annotations: data linking

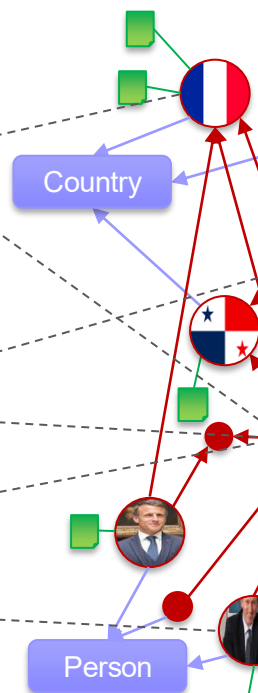
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Document file

Annotation file



```
{
  "annotation_sets": {
    "sections": {},
    "sentences": {},
    "entities": {
      "annotations": [
        {
          "type": "PERSON",
          "start": 0,
          "end": 23,
          "id": 0,
          "features": {
            "ner": { "type": "PERSON" },
            "linking": {
              "encoding": "",
              "source": "blink_biencoder",
              "top_candidate": {
                "id": 213062,
                "indexer": 10,
                "wikipedia_id": 534366,
                "title": "Barack Obama",
                "score": 77.57928466796875 },
                "candidates": [],
                "nil_score": 0.43745980781641175,
                "is_nil": true },
              "cluster": 1 }}}],
          "text": "Barack Hussein Obama II (born August 4, 1961) is an ...",
          "features": {}
        }
      ]
    }
  }
```


~ Semantic Data Enrichment

Annotations: data linking

BROWSE BY COUNTRY FRANCE

ENTITY	JURISDICTION	LINKED TO	DATA FROM
OBOLT GLOBAL LTD	British Virgin Islands	France	Panama Papers
FAIRWINDS SAILING S.A.	British Virgin Islands	France	Panama Papers
SIENNA HOLDINGS INC.	Panama	France	Panama Papers
PROCESOS E INGENIERIA S.A.	Panama	France	Panama Papers
WEBCO INTERNATIONAL S.A.	Bahamas	France	Panama Papers
INTERPETROLEUM LTD.	Bahamas	France	Panama Papers
UNICONSTRUCT INC.	Panama	France	Panama Papers
MADINA AL MINWORA AGENCY CORPORATION	Panama	France	Panama Papers
CARLTON PROPERTIES LIMITED	Bahamas	France	Panama Papers
BELROS HOLDINGS LTD.	Bahamas	France	Panama Papers

FRANCE - PANAMA

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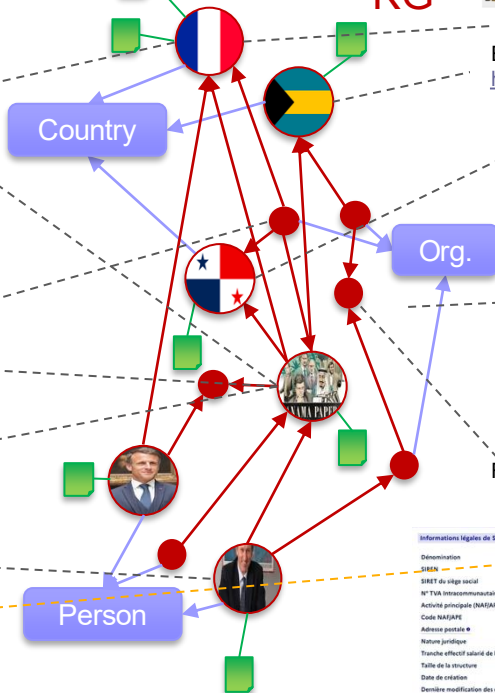
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Par Jérémie Baruch

Publié le 21 juin 2018 à 10h54, modifié le 22 juin 2018 à 07h12 - Lecture 3 min.

... Sienna ...

Background KG



Entities in OffshoreLeaks linked to France

<https://offshoreleaks.icij.org/search?c=FRA&cat=0>

Foundations firms 'offshore' customers through banks

Headquarters	Bank	Number of foundations
Luxembourg	Experta Corporate & Trust Services (100% subsidiary of BGL)	1,659
Luxembourg	Banque J. Safra Sarasin - Luxembourg S.A.	963
United Kingdom	Street Suisse Channel Islands Limited	918
Monaco	HSBC Private Bank (Monaco) S.A.	778
Switzerland	HSBC Private Bank (Switzerland) S.A.	733
Switzerland	UBS AG (subsidiary Rue du Rhône in Geneva)	579
United Kingdom	Coutts & Co Trustees (Jersey) Limited	487
Jersey		
Luxembourg	Société Générale Bank & Trust Luxembourg	465
Luxembourg	Landbank Luxembourg S.A.	404
United Kingdom	Barclays Bank Trust (Jersey) Limited	378
Spain	Banco Santander	119
Spain	BBVA	19

Foundations firms 'offshore' customers through banks in Wikipedia

Informations légales de SIENNA REAL ESTATE HOLDING FRANCE

Dénomination	492 220 553
SIENNA	492 220 553 00027
SIRET du siège social	519 492 220 553
N° TVA Intracommunautaire	Façon de placement et entités financières similaires
Activité principale (NAIS/AFPE)	64 30Z
Code NAF/AFPE	18 RUE DE COURCELLES, 75008 PARIS 8
Adresse postale	SAS, société par actions simplifiée
Nature juridique	Unité non employeur
Taux effectif d'impôt de la structure	Prévue au Régime des Entreprises (PME), en 2020
Taille de la structure	2009/2008
Date de création	06/09/2002
Dernière modification des données Insee	

Justification d'existence

- Téléchargez l'extrait RNE (E)
- Avis de situation Insee (E)

Company data from .data.gouv.fr

<https://annuaire-entreprises.data.gouv.fr/entreprise/sienna-real-estate-rise/france-492220553>

Named Entity Recognition

Named Entity Linking

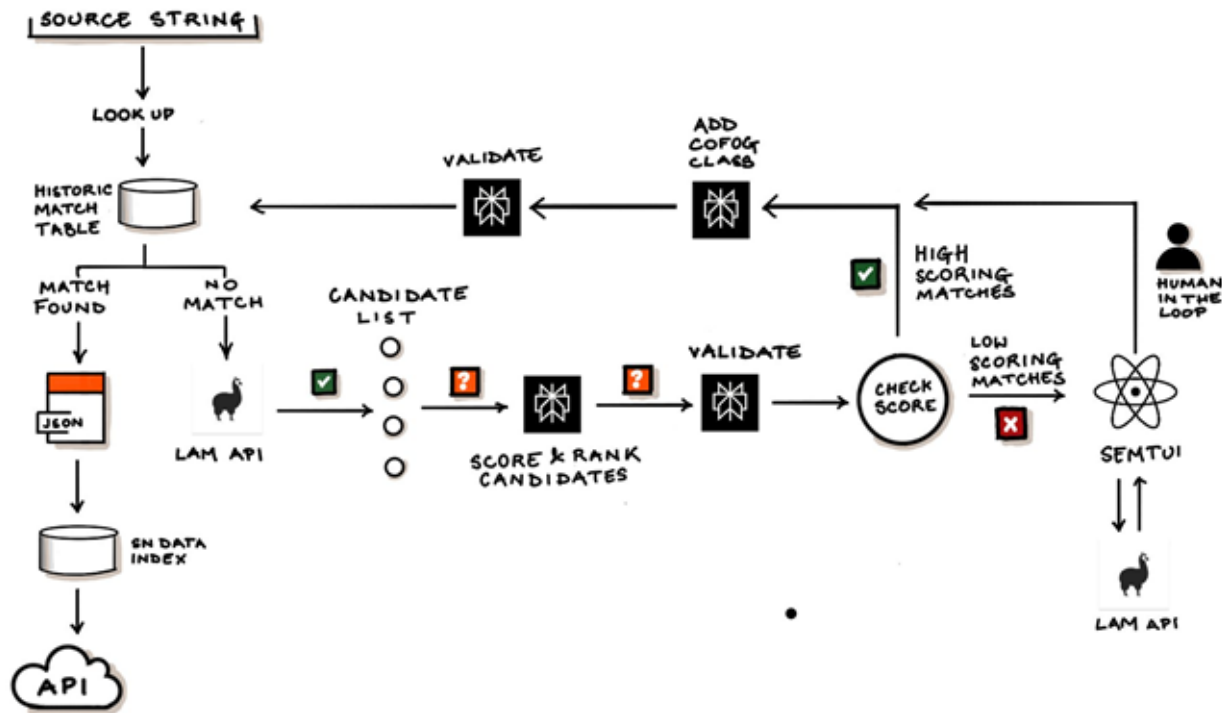
NIL Prediction

Clustering

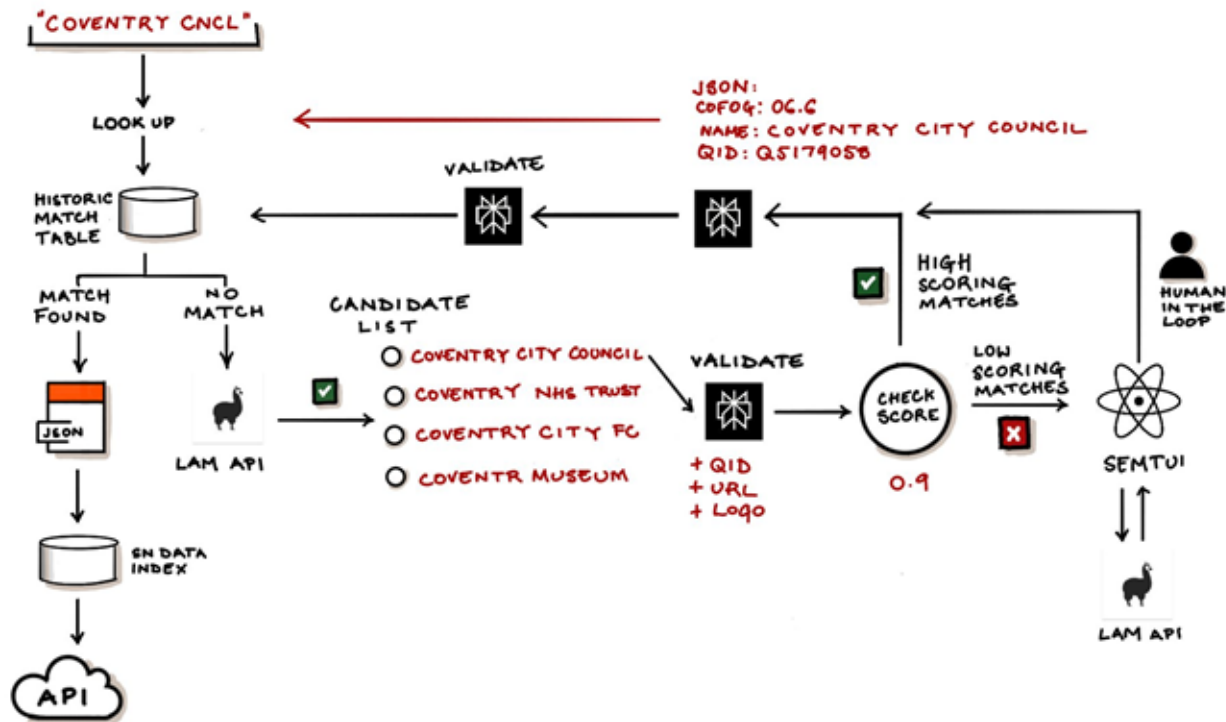
Spend Network:

**Easy access and analysis for
the world's \$13trn government
contracting market.**

Enrichment - Buyers



Enrichment - Buyers



Key results

Notes on results

- > Match rates impacted by Wikidata cover
- > Pre-validation match was 7% now 42%
- > 38% of EU tenders matched



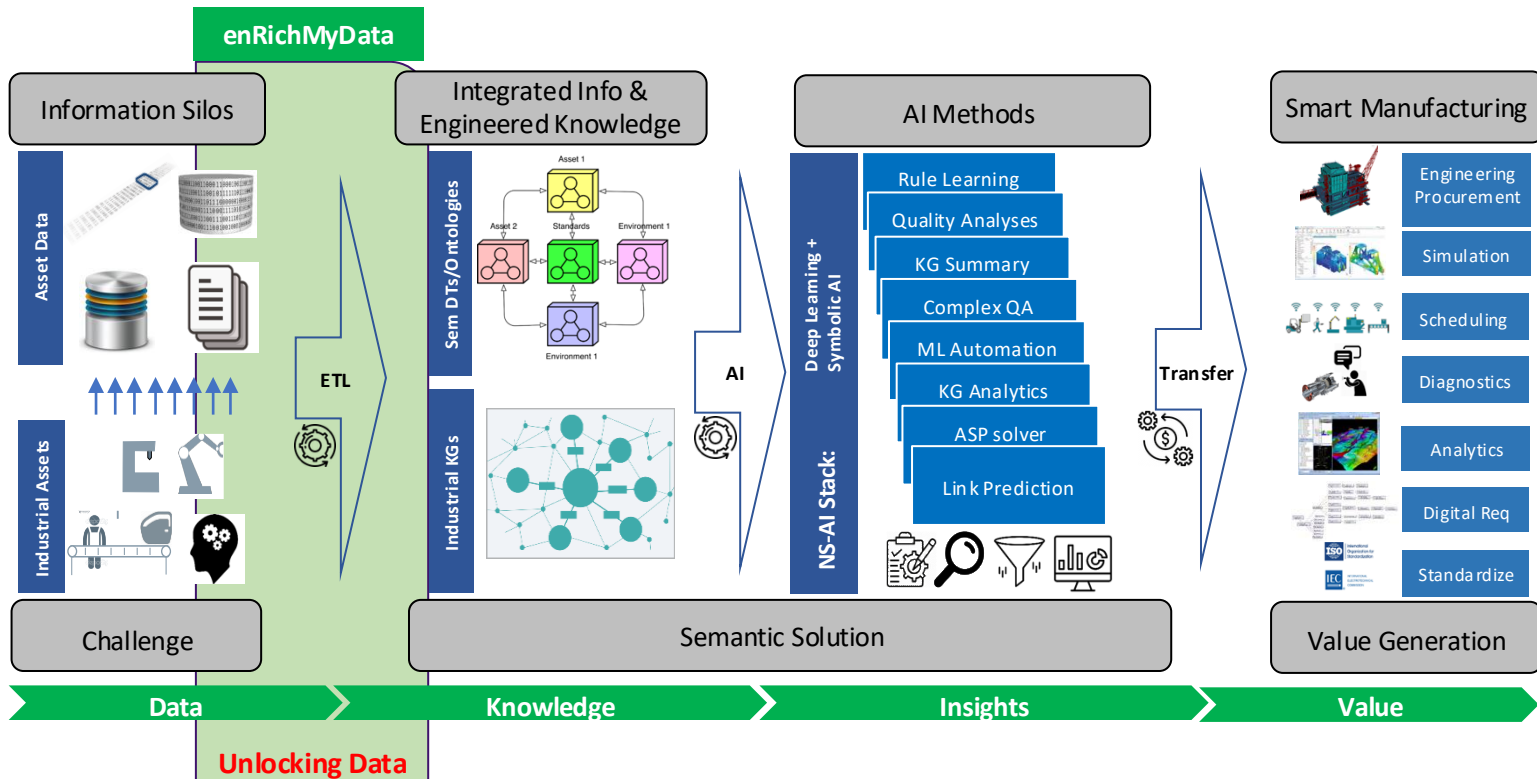
BC2: AI-Based Welding Analytics (AIWela)

Dr. Cuong Xuan Chu (Bosch Corporate Research)

Final Review

Dec 02, 2025

BOSCH –Industrial KGs & Semantics Models



EventRegistry.org: Event Tracking at Massive Scale with KGs



KDD 2020 Tutorial | KDD Tutorial 1 - Introduction.pptx | Event Registry - searching for

eventregistry.org/intelligence?tab=items&searchMode=simple&type=events&conditions=2--1-1-Giorgia%20Meloni%20&

General (enRichM... | Apple | Yahoo! | Around Semantics | Math | Movies | Socio-economics | Google Maps | YouTube

Giorgia Meloni

ARTICLES EVENTS

Filters: Locations Sources Categories Last month Any language Misc.

List of events (2,416 results found)

Note: Since the simple search mode was used, the results might mention just a subset of the entered keywords. To change the search mode, click the icon next to the Search button.

VIEW: List SORT BY: Relevance

EVENT

Pilgrims to Mussolini's birthplace pray that new PM will resurrect a far-right Italy

WHEN: Fri, October 21, 2022

WHERE: Rome, Italy

ARTICLES: 452

VIRALITY: 72

SENTIMENT: **

Giorgia Meloni Italy Brothers of Italy Coalition government Silvio Berlusconi Far-right politics Sergio Mattarella Forza Italia

Right-wing politics Matteo Salvini Ukraine Rome, Italy Russia Mario Draghi Prime Minister of the United Kingdom Cabinet (government)

In Predappio, supporters celebrate victory of their first female prime minister Giorgia Meloni, leader of a party with neo-fascist origins Dressed in... (The Guardian)

EVENT

Macron a rencontré Meloni à Rome, avec qui il promet "dialogue et ambition"

WHEN: Fri, October 21, 2022

WHERE: Rome, Italy

ARTICLES: 214

VIRALITY: 73

SENTIMENT: N/A

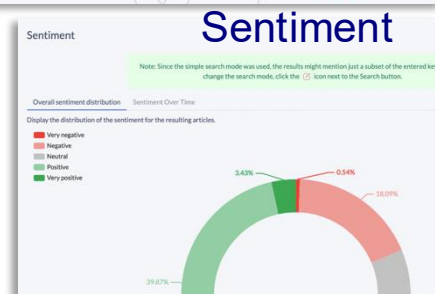
Giorgia Meloni Government Italy Matteo Salvini Prime minister Silvio Berlusconi Rome, Italy Sergio Mattarella Coalition government

Forza Italia Mario Draghi Benito Mussolini President of France Quirinal Palace Populism Antonio Tajani Euroscepticism Europe

INTERNATIONAL - À peine entrée en fonction, la nouvelle Première ministre italienne Giorgia Meloni a fait ses débuts sur la scène internationale ce di... (Le Huffington Post)

Show all

Events
clusters of news about a same topic, based on semantic annotations with entities, concepts, time, place



Link to web app (restricted access): <https://eventregistry.org/>

Features & Impact on Applications

Impact of these features depend on applications and technical implementation choices (e.g., RDF vs. others)

- Factuality and reasoning: question answering
- Natural language processing (and NLP)
- Graphs
- Interoperability: simple triple-based model + standard formats + shared vocabularies
- Data integration: linked data

KGs for LLMs

Knowledge Graphs as a source of trust for LLM-powered enterprise question answering

Juan Sequeda*, Dean Allemang, Bryon Jacob

data.world AI Lab, Austin, TX, USA

GPT-4 powered question answering system that answers enterprise natural language questions

- **Knowledge Graph representation of an enterprise SQL database** → 3x more accurate than SQL-only database schema.
- Using an **ontology** to identify semantic errors in AI-generated SPARQL queries + **LLM repairs** → 4x accuracy vs. no Knowledge Graph

LLMs for QA on Graphs

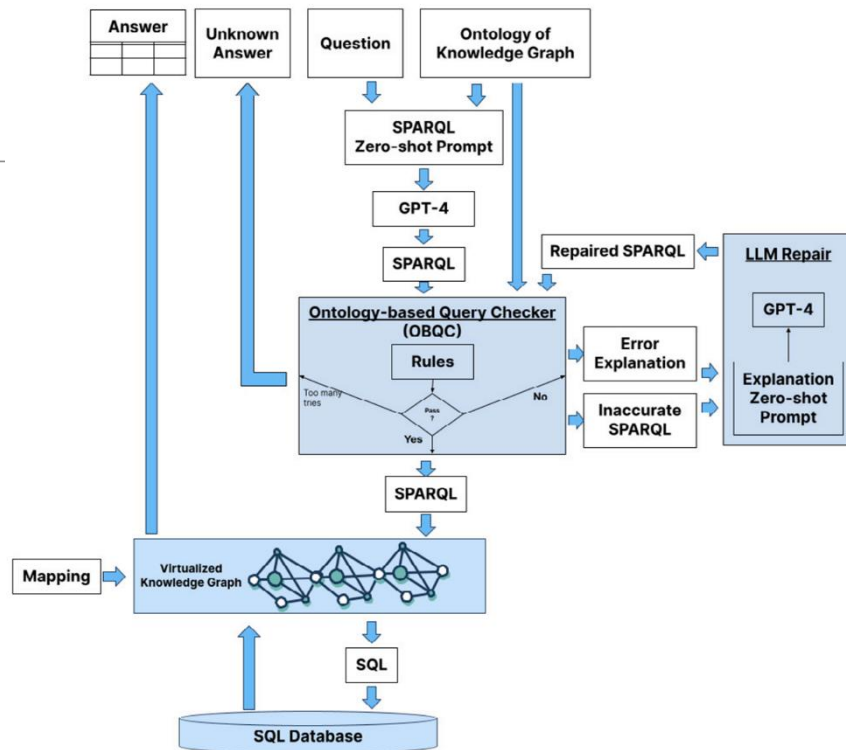


Fig. 1. Overview of our approach.

Risks of GenAI and Factual Content Generation

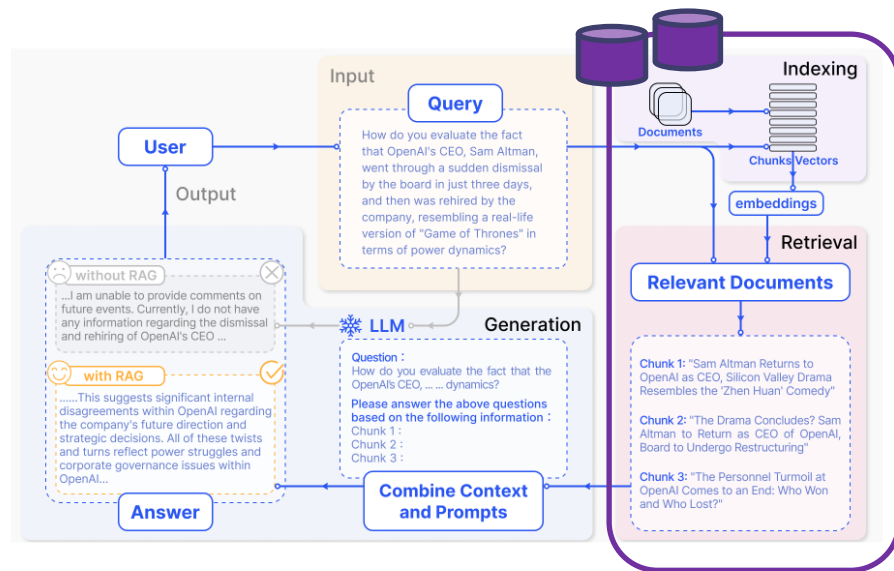
■ Limitations of Large Language Models (LLMs)

- Output based on opaque parametric knowledge acquired during training
 - Hard to update
 - Hard to fix / correct
- Risks: hallucinations, offensive or dangerous content, costs, etc.
 - Affected also by context size, e.g., very large document

■ A task or question may require access to data that LLMs cannot access

- Because of: timeliness, privacy, security, size
- Consisting of trusted data sources, incl. factual information
 - documents, images, structured databases, graphs

>> Retrieval-augmented Generation (RAG)



The model retrieves data from one or more sources and uses the retrieved data for generation

RAG@LinkedIn: Knowledge Graph + LLMs for CRM

Xu, Z., Cruz, M. J., Guevara, M., Wang, T., Deshpande, M., Wang, X., & Li, Z. (2024). **Retrieval-Augmented Generation with Knowledge Graphs for Customer Service Question Answering**. arXiv preprint <https://arxiv.org/pdf/2404.17723>.

Performance improvement

Content
retrieval

	MRR	Recall@K		NDCG@K	
		K=1	K=3	K=1	K=3
Baseline	0.522	0.400	0.640	0.400	0.520
Experiment	0.927	0.860	1.000	0.860	0.946

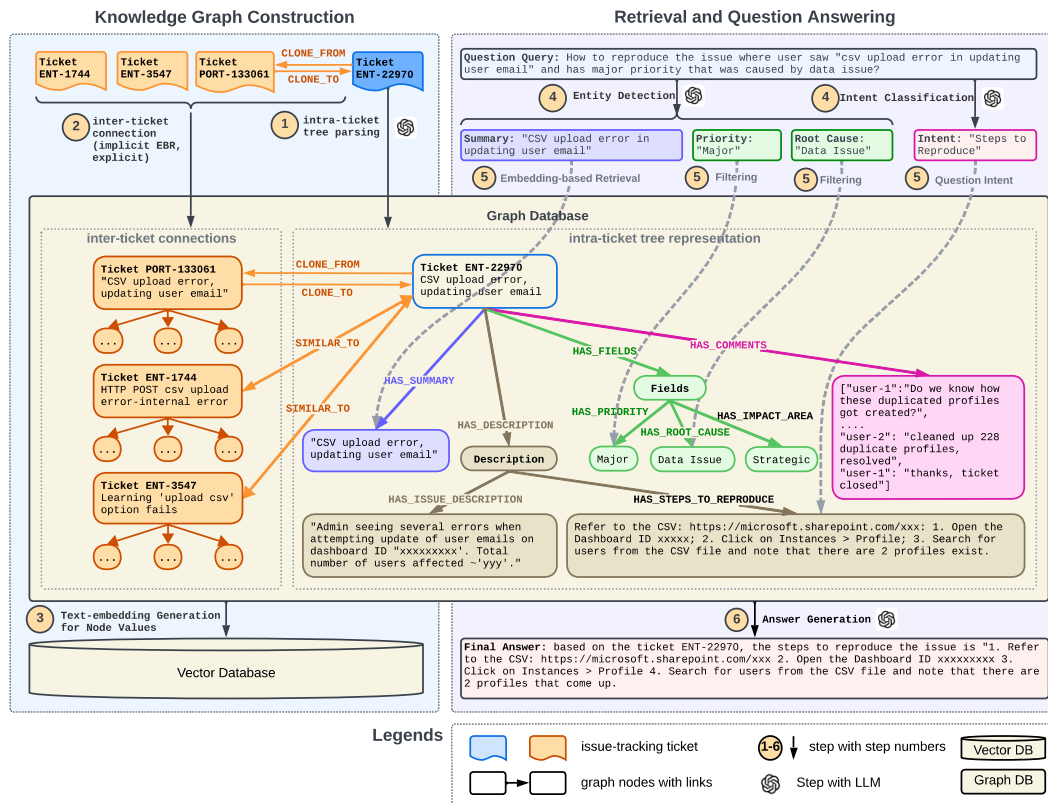
Answer
quality

	BLEU	METEOR	ROUGE
Baseline	0.057	0.279	0.183
Experiment	0.377	0.613	0.546

Impatto e valore

Business
impact: reduced
time for user
support

Group	Mean	P50	P90
Tool Not Used	40 Hours	7 Hours	87 Hours
Tool Used	15 hours	5 hours	47 hours



Knowledge Graphs for Data Enrichment & Analytics

- Knowledge Graphs & Semantics In-use for Investigative Applications:
 - Szekely & al. Building and Using a Knowledge Graph to Combat Human Trafficking. ISWC 2015. https://doi.org/10.1007/978-3-319-25010-6_12
- EW-Shopp / enRichMyData Projects
 - Website: <https://www.ew-shopp.eu/> - Data blog: <https://www.ew-shopp.eu/data-blog/>
 - Website: <https://enrichmydata.eu/> - Data blog: <https://enrichmydata.eu/blog/>
 - Semantics to support digital marketing campaigns (paper): https://doi.org/10.1007/978-3-030-30796-7_22
- Semantic enrichment of tabular data:
 - Semantic Data Enrichment for Tabular Data (tutorial @ESWC2024): <https://enrichmydata.github.io/eswc2024-tutorial/>